Square-Energy Asymmetry from Packing at Most Two Cycles

Abstract

For a finite simple graph G, let $s^+(G)$ and $s^-(G)$ be the sums of the squares of its positive and negative adjacency eigenvalues. We prove that $s^+(G) > s^-(G)$ whenever G contains a cycle, every cycle of G has length 3 modulo 4, and the maximum size of a collection of pairwise vertex-disjoint cycles is at most two. If every cycle has length 1 modulo 4, the inequality is reversed. The proof groups the Sachs expansion by collections of vertex-disjoint cycles. On the imaginary axis each cycle contributes the phase $-2\mathrm{i}^{-\ell}$, while the remaining factor is a signless matching polynomial with nonnegative coefficients and positive value on the positive axis. Packing number at most two then puts the normalized characteristic polynomial strictly in one open half-plane. A Coulson identity converts this fixed argument sign into the claimed square-energy inequality. Applications include a strict form of the positive square-energy conjecture for every triangular block graph and for block graphs of cyclomatic numbers two through five.

1 Introduction

Let G be a finite simple graph with adjacency eigenvalues $\lambda_1, \ldots, \lambda_n$. Its positive and negative square energies are

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \qquad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2.$$

Since $\operatorname{tr} A(G)^2 = 2|E(G)|$,

$$s^+(G) + s^-(G) = 2|E(G)|. \tag{1}$$

These quantities were introduced in connection with spectral bounds for the chromatic number by Elphick, Farber, Goldberg, and Wocjan [5]; subsequent structural and asymmetry results appear in [3, 6, 11]. The conjecture that every connected n-vertex graph satisfies $\min\{s^+, s^-\} \geq n - 1$ has recently been proved by Liu, Tang, and Zhang [8]. Akbari, Kumar, Mohar, Pragada, and Zhang proposed the stronger conjecture that a connected graph with at least $n + 1$ edges satisfies $s^+(G) \geq n$ [1, Conjecture 1.2].

For a graph G, write $\nu_\circ(G)$ for the maximum cardinality of a collection of pairwise vertex-disjoint cycles. Our main result is the following sign theorem. Connectivity is not required.

Theorem 1.1. *Let G be a finite simple graph that contains a cycle and satisfies $\nu_\circ(G) \leq 2$.*

(i) *If every cycle of G has length congruent to $3 \pmod 4$, then $s^+(G) > s^-(G)$.*

(ii) *If every cycle of G has length congruent to $1 \pmod 4$, then $s^+(G) < s^-(G)$.*

Ning and Zeng proved the corresponding statement for an odd unicyclic graph [9]. Their argument also uses a matching polynomial and a Coulson-type identity. Their result is the one-cycle case of the sign assertion above; the new issue here is that two-cycle terms can make the real part of the normalized characteristic polynomial negative. They remain real, however, so the open-half-plane argument survives. Here the full Sachs expansion is grouped by vertex-disjoint cycle collections; the terms with two cycles are real, whereas the terms with one cycle have a common nonzero imaginary sign. This is exactly where the hypothesis $\nu_\circ(G) \le 2$ enters. The reverse assertion in Theorem 1.1(ii) is included as the phase-reversed companion statement. We do not make a novelty or priority claim on the basis of a negative literature search.

Section 2 gives a self-contained proof, including the required Coulson identity and a careful continuous-argument step. Section 3 records the cactus structure forced by the cycle congruence, and Section 4 gives applications, including a cycle-territory decomposition and block graphs through cyclomatic number five. We also use the induced-subgraph superadditivity theorem of Akbari, Kumar, Mohar, and Pragada [2]: if $H_1, \dots, H_r$ are pairwise disjoint induced subgraphs of G, then

$$s^+(G) \ge \sum_{j=1}^{r} s^+(H_j). \tag{2}$$

2 Sachs expansion on the imaginary axis

For a graph H of order h, let $m_j(H)$ denote its number of j-edge matchings and define the following normalization of the matching polynomial; see Godsil and Gutman [7]:

$$Z_H(t) = \sum_{j=0}^{\lfloor h/2 \rfloor} m_j(H) t^{h-2j}. \tag{3}$$

We use $Z_\varnothing(t) = 1$. Thus $Z_H(t) > 0$ for every $t > 0$. Equivalently, if

$$\mu_H(x) = \sum_j (-1)^j m_j(H) x^{h-2j}$$

is the matching polynomial, then

$$\mathrm{i}^{-h} \mu_H(\mathrm{i}t) = Z_H(t). \tag{4}$$

Let $\mathfrak{C}(G)$ be the set of all collections $\mathcal{C}$ of pairwise vertex-disjoint cycles of G, including the empty collection. Put $V(\mathcal{C}) = \bigcup_{C \in \mathcal{C}} V(C)$.

Lemma 2.1 (Sachs expansion grouped by cycles)**.** *If G has order n and characteristic polynomial $\phi_G(x) = \det(xI - A(G))$, then*

$$\phi_G(x) = \sum_{\mathcal{C} \in \mathfrak{C}(G)} (-2)^{|\mathcal{C}|} \mu_{G - V(\mathcal{C})}(x). \tag{5}$$

Consequently, for $t > 0$,

$$\Psi_G(t) := \mathrm{i}^{-n} \phi_G(\mathrm{i}t) = \sum_{\mathcal{C} \in \mathfrak{C}(G)} \left(\prod_{C \in \mathcal{C}} q_{|C|} \right) Z_{G - V(\mathcal{C})}(t), \qquad q_\ell = -2\mathrm{i}^{-\ell}. \tag{6}$$

Proof. Sachs' formula [10] sums over elementary subgraphs S, whose components are edges and cycles:

$$\phi_G(x) = \sum_S (-1)^{c(S)} 2^{c_\circ(S)} x^{n - |V(S)|},$$

where $c(S)$ and $c_\circ(S)$ are the numbers of components and cycle components of S. Fix its cycle components as a collection $\mathcal{C}$. The other components form a matching of some size j in $G - V(\mathcal{C})$. The corresponding sign and weight are $(-1)^{|\mathcal{C}|+j}2^{|\mathcal{C}|}$. Summing over those matchings gives $(-2)^{|\mathcal{C}|}\mu_{G-V(\mathcal{C})}(x)$, proving (5).

For a fixed $\mathcal{C}$, set $L = \sum_{C\in\mathcal{C}} |C|$. Substitution of $\mathrm{i}t$ and (4) give

$$\mathrm{i}^{-n}(-2)^{|\mathcal{C}|}\mu_{G-V(\mathcal{C})}(\mathrm{i}t) = (-2)^{|\mathcal{C}|}\mathrm{i}^{-L}Z_{G-V(\mathcal{C})}(t),$$

which factors into one $q_{|C|}$ for each $C \in \mathcal{C}$. $\square$

The next identity is a signed square-energy version of the classical Coulson integral formula. We include the derivation to fix both the sign and the branch convention; compare [4, 9].

Lemma 2.2 (Coulson identity)**.** *For $t > 0$, define*

$$\Theta_G(t) = \sum_{j=1}^{n} \arctan\left(\frac{\lambda_j}{t}\right), \qquad -\frac{\pi}{2} < \arctan x < \frac{\pi}{2}. \tag{7}$$

Then the integral below is absolutely convergent and

$$s^+(G) - s^-(G) = -\frac{4}{\pi}\int_0^\infty t\Theta_G(t)\,dt. \tag{8}$$

Proof. For every real λ,

$$\lambda|\lambda| = \frac{2}{\pi}\int_0^\infty \frac{\lambda^3}{\lambda^2 + t^2}\,dt.$$

The scalar integral is absolutely convergent. Since there are finitely many eigenvalues and $\sum_j \lambda_j = \operatorname{tr} A(G) = 0$,

$$s^+ - s^- = \frac{2}{\pi}\int_0^\infty \sum_j \frac{\lambda_j^3}{\lambda_j^2 + t^2}\,dt = \frac{2}{\pi}\int_0^\infty t^2\Theta_G'(t)\,dt.$$

Indeed, $\Theta_G'(t) = -\sum_j \lambda_j/(\lambda_j^2 + t^2)$ and $\sum_j \lambda_j^3/(\lambda_j^2 + t^2) = -t^2\sum_j \lambda_j/(\lambda_j^2 + t^2)$.

Integrate by parts first on $[\varepsilon, R]$. At zero, $|\Theta_G(t)| \le n\pi/2$, so $t^2\Theta_G(t) \to 0$. At infinity, Taylor expansion and $\sum_j \lambda_j = 0$ give $\Theta_G(t) = O(t^{-3})$, so again $t^2\Theta_G(t) \to 0$. The same estimates show that $t\Theta_G(t)$ is absolutely integrable: it is $O(t)$ at zero and $O(t^{-2})$ at infinity. Letting $\varepsilon \downarrow 0$ and $R \to \infty$ yields (8). $\square$

Proof of Theorem 1.1. Suppose first that every cycle length is $3 \pmod 4$. For such a length,

$$q_\ell = -2\mathrm{i}^{-\ell} = -2\mathrm{i}.$$

Because $\nu_\circ(G) \le 2$, expansion (6) contains only collections of zero, one, or two cycles. The zero-cycle term is real; each two-cycle term is real because $(-2\mathrm{i})^2 = -4$; and every one-cycle term has imaginary part $-2Z_{G-V(C)}(t) < 0$. Since G contains a cycle, it follows that

$$\operatorname{Im}\Psi_G(t) = -2\sum_{C \text{ a cycle of } G} Z_{G-V(C)}(t) < 0 \qquad (t > 0). \tag{9}$$

We now justify the argument without an implicit branch choice. Directly from the eigenvalue factorization,

$$\Psi_G(t) = \mathrm{i}^{-n}\prod_j(\mathrm{i}t - \lambda_j) = \prod_j(t + \mathrm{i}\lambda_j). \tag{10}$$

Thus $\Theta_G(t)$ is a continuous argument of $\Psi_G(t)$. By (9), the curve $\Psi_G((0,\infty))$ lies strictly in the open lower half-plane. Its principal argument $\alpha(t) = \operatorname{Arg}\Psi_G(t) \in (-\pi, 0)$ is therefore continuous. Moreover, $\Theta_G(t) \to 0$ as $t \to \infty$, while $\Psi_G(t) = t^n(1 + o(1))$ and (9) imply $\alpha(t) \to 0$ from below. The continuous function $\Theta_G(t) - \alpha(t)$ takes values in $2\pi\mathbb{Z}$, hence is constant; its limit at infinity is zero. Therefore $\Theta_G(t) = \alpha(t) < 0$ for every $t > 0$. In particular the integral in (8) is strictly negative, and $s^+(G) - s^-(G) > 0$.

If every cycle length is $1 \pmod 4$, then $q_\ell = 2\mathrm{i}$. The two-cycle terms are again real, while

$$\operatorname{Im}\Psi_G(t) = 2 \sum_{C \text{ a cycle of } G} Z_{G-V(C)}(t) > 0.$$

The same argument in the open upper half-plane shows that $\Theta_G(t) \in (0, \pi)$ for every $t > 0$. Lemma 2.2 now gives $s^+(G) - s^-(G) < 0$. □

Remark 2.3. The proof applies verbatim to disconnected graphs. Equivalently, ϕ_G and Ψ_G factor over components and the square energies are additive. Forest components contribute zero asymmetry, while the common cycle phase controls all cyclic components under the global packing hypothesis.

3 The structure forced by one cycle phase

We recall that a cactus is a graph in which every block is either a single edge or a cycle. Isolated vertices are allowed here.

Lemma 3.1. *If every cycle of a finite simple graph G has length congruent to* $3 \pmod 4$*, then every cyclic block of G is one of those cycles. In particular, G is a cactus. The same conclusion holds with* $3 \pmod 4$ *replaced by* $1 \pmod 4$*.*

Proof. A theta graph consists of three internally vertex-disjoint paths between the same two distinct vertices. If their lengths are a, b, c, its three cycles have lengths $a + b$, $a + c$, and $b + c$. These three lengths cannot all be congruent to the same odd residue $r \pmod 4$, since

$$2a = (a + b) + (a + c) - (b + c) \equiv r \pmod 4,$$

whose left side is even and right side is odd.

It remains to recall why a block that is neither an edge nor a cycle contains a theta subdivision. Let B be such a block. It is 2-connected, and it contains a cycle C. If B has an edge not on C, take a component K of $B - V(C)$ incident with C, or a chord of C if there is no such component. In the former case K has at least two distinct attachment vertices u, v on C: a unique attachment would be a cut vertex of the 2-connected graph B. Since K is connected, a path in K between neighbors of two such attachments, together with its two attachment edges, gives a u–v path internally disjoint from C. In the chord case the chord is such a path. Together with the two u–v arcs of C, this is a theta subdivision, a contradiction. Hence every nontrivial block is an edge or a cycle. □

The lemma also avoids any need for an unproved classification of how triangles can intersect: under the cycle-congruence hypothesis the full block structure follows from the elementary theta obstruction.

4 Applications

For a graph G with $\kappa(G)$ components, its cyclomatic number is

$$\beta(G) = |E(G)| - |V(G)| + \kappa(G).$$

A block graph is a graph every block of which is a clique.

We first record a decomposition that will also remove any global bound on the number of triangles in a triangular block graph. The priority order in the next lemma is fixed once and for all.

Lemma 4.1 (Cycle territories)**.** *Let G be a connected graph, and let $C_1, \dots, C_k$, where $k \geq 1$, be a maximum-cardinality collection of pairwise vertex-disjoint cycles. Assign each vertex v to the index i for which*

$$\big(d_G(v, V(C_i)), i\big)$$

is lexicographically least, and let G_i be the subgraph induced by the vertices assigned to i. Then the sets $V(G_1), \dots, V(G_k)$ partition $V(G)$, and every G_i is connected, contains C_i, and has cycle packing number exactly one.

Proof. The assignment plainly gives a partition. Since the cycles in the chosen packing are vertex-disjoint, every vertex of C_i has distance zero from C_i and positive distance from every C_j with $j \neq i$. Hence $C_i \subseteq G_i$.

We prove connectivity with the integer distances and the tie-breaking rule made explicit. Let $v \in V(G_i) \setminus V(C_i)$, put $a = d_G(v, V(C_i))$, and choose a neighbor u of v on a shortest v–C_i path. Thus $d_G(u, V(C_i)) = a - 1$. If $j < i$, then priority would favor j in a tie, so the assignment of v to i implies

$$a < d_G(v, V(C_j)).$$

Both quantities are integers, and the distance function changes by at most one across an edge. Consequently

$$d_G(u, V(C_j)) \geq d_G(v, V(C_j)) - 1 \geq a > a - 1 = d_G(u, V(C_i)).$$

If instead $j > i$, assignment of v to i gives $a \leq d_G(v, V(C_j))$, and hence

$$d_G(u, V(C_j)) \geq d_G(v, V(C_j)) - 1 \geq a - 1 = d_G(u, V(C_i)).$$

In this case equality still assigns u to i, because $i < j$. Thus the chosen predecessor u belongs to G_i. Iterating reaches C_i entirely inside G_i, proving that G_i is connected.

Finally, G_i contains a cycle, so $\nu_\circ(G_i) \geq 1$. If some G_i contained two vertex-disjoint cycles, those two cycles together with C_j for every $j \neq i$ would be $k+1$ pairwise vertex-disjoint cycles in G, because the territories are disjoint. This contradicts the *maximum* cardinality of the chosen packing. Therefore $\nu_\circ(G_i) = 1$. □

Corollary 4.2 (Triangular block graphs)**.** *Every connected triangular block graph G of order n that contains a triangle satisfies*

$$s^+(G) > n.$$

Here triangular means that every cyclic block is a triangle; bridge blocks and arbitrary attached trees are allowed.

Proof. Apply Lemma 4.1 to a maximum, not merely maximal, cycle packing and write $n_i = |V(G_i)|$ and $m_i = |E(G_i)|$. Each connected induced territory contains a triangle, all of its cycles are triangles, and its cycle packing number is one. Theorem 1.1(i) therefore gives $s^+(G_i) > s^-(G_i)$. Since $s^+(G_i) + s^-(G_i) = 2m_i$, it follows that

$$s^+(G_i) > m_i \geq n_i,$$

where the last inequality holds because G_i is connected and cyclic. The territories are pairwise disjoint induced subgraphs, so (2) yields

$$s^+(G) \geq \sum_{i=1}^{k} s^+(G_i) > \sum_{i=1}^{k} n_i = n.$$

Notice that this argument does not assert $s^+(G) > |E(G)|$. $\square$

Corollary 4.3 (Bicyclic block graphs)**.** *Let G be a connected block graph with n vertices and cyclomatic number 2. Then G has exactly two blocks isomorphic to K_3, all its other blocks are copies of K_2, and*

$$s^+(G) > s^-(G), \qquad s^+(G) > |E(G)| = n + 1 > s^-(G).$$

In particular, Conjecture 1.2 of Akbari–Kumar–Mohar–Pragada–Zhang [1] holds strictly, and in the stronger form $s^+(G) > n + 1$, for every bicyclic block graph.

Proof. Cyclomatic number is additive over the blocks of a connected graph. A clique block K_r contributes

$$|E(K_r)| - |V(K_r)| + 1 = \binom{r}{2} - r + 1 = \frac{(r-1)(r-2)}{2}.$$

This contribution is 0 for $r = 2$, 1 for $r = 3$, and at least 3 for $r \geq 4$. Since the total is 2, there are exactly two K_3 blocks and all other blocks are K_2's. Thus the only cycles are the two triangles, so their vertex-disjoint packing number is at most two. Theorem 1.1(i) gives $s^+ > s^-$. Finally $|E(G)| = n - 1 + \beta(G) = n + 1$, and (1) says that the average of s^+ and s^- is $|E(G)|$. The displayed strict inequalities follow. $\square$

The next application goes one cycle rank farther. Its packing-three case is handled by cutting bridges rather than by extending the half-plane argument: three-cycle Sachs terms have the opposite imaginary sign.

Proposition 4.4 (Tricyclic block graphs)**.** *Let G be a connected block graph with n vertices and cyclomatic number 3. Then*

$$s^+(G) > n.$$

Thus Conjecture 1.2 of Akbari–Kumar–Mohar–Pragada–Zhang holds strictly for every tricyclic block graph.

Proof. As in the proof of Corollary 4.3, the positive contributions of clique blocks to the cyclomatic number are $1, 3, 6, \ldots$. Hence either G has one K_4 block and all other blocks are bridges, or it has three K_3 blocks and all other blocks are bridges.

Suppose first that the unique cyclic block is K_4. Every cycle lies in this block, and no two cycles are vertex-disjoint. In the normalized Sachs expansion, a triangle has phase $q_3 = -2\mathrm{i}$, while a 4-cycle has real phase $q_4 = -2$. Therefore

$$\operatorname{Im} \Psi_G(t) = -2 \sum_{C \subseteq K_4,\ C \cong C_3} Z_{G - V(C)}(t) < 0 \qquad (t > 0).$$

The continuous-argument proof of Theorem 1.1 applies verbatim and gives $s^+(G) > s^-(G)$. Since $|E(G)| = n + 2$, this yields $s^+(G) > n + 2$.

Now suppose that the cyclic blocks are three triangles. If their vertex-disjoint packing number is at most two, Theorem 1.1(i) again gives $s^+(G) > s^-(G)$ and hence $s^+(G) > n + 2$.

It remains that the three triangle blocks are pairwise vertex-disjoint. Contract each triangle block to one marked vertex. Since every remaining block is a bridge, the resulting graph is a tree with three distinct marked vertices. Two edges of the minimal subtree joining the marks can be deleted so that the resulting three components contain one mark each. Each selected tree edge corresponds to an actual bridge of G. Deleting those two bridges therefore partitions $V(G)$ into three sets inducing connected graphs G_1, G_2, G_3, each with exactly one triangle and otherwise only bridge blocks. Thus each G_j is a triangular unicyclic graph. Ning and Zeng [9] give

$$s^+(G_j) > |V(G_j)|.$$

Applying (2) to this induced vertex partition gives

$$s^+(G) \geq \sum_{j=1}^{3} s^+(G_j) > \sum_{j=1}^{3} |V(G_j)| = n.$$

This completes all cases. □

The tetracyclic case requires one additional configuration. In that configuration three disjoint triangle blocks meet a fourth triangle block, and the one-cycle and three-cycle terms in the Sachs expansion must be compared quantitatively.

Proposition 4.5 (Tetracyclic block graphs)**.** *Let G be a connected block graph with n vertices and cyclomatic number 4. Then*

$$s^+(G) > n.$$

Thus Conjecture 1.2 of Akbari–Kumar–Mohar–Pragada–Zhang holds strictly for every tetracyclic block graph.

Proof. The contribution of a clique block K_r to the cyclomatic number is $(r-1)(r-2)/2$. Consequently, the cyclic blocks of G are either one K_4 and one K_3, or four copies of K_3; all remaining blocks are bridges. Also

$$|E(G)| = n - 1 + \beta(G) = n + 3. \tag{11}$$

First suppose that the cyclic blocks are a K_4 and a K_3. If they are vertex-disjoint, the path between them in the block–cut tree contains a bridge block. Deleting the corresponding bridge partitions $V(G)$ into two sets inducing connected block graphs G_1, G_2, one containing the K_4 and the other the K_3. Proposition 4.4 and the odd-unicyclic result of Ning and Zeng give

$$s^+(G) \geq s^+(G_1) + s^+(G_2) > |V(G_1)| + |V(G_2)| = n.$$

Here and below we use induced-subgraph superadditivity (2) for the vertex partition produced by deleting bridges.

Otherwise the two clique blocks share their unique possible common vertex, which is a cut vertex. We apply the grouped Sachs expansion directly. Every triangle has phase $-2\mathrm{i}$, while every 4-cycle has the real phase -2. The triangle of the K_4 avoiding the common cut vertex is vertex-disjoint from the external K_3, but that two-triangle term is real. Every 4-cycle of the K_4 contains the cut vertex, so no 4-cycle is disjoint from the external triangle. There are no other collections that can contribute to the imaginary part. Hence

$$\operatorname{Im} \Psi_G(t) = -2 \sum_{T \text{ a triangle of } G} Z_{G-V(T)}(t) < 0 \qquad (t > 0).$$

The continuous-argument and Coulson steps in the proof of Theorem 1.1 yield $s^+(G) > s^-(G)$, and then (1) and (11) give $s^+(G) > |E(G)| = n + 3 > n$.

It remains to consider four triangle blocks. Write p for their maximum vertex-disjoint packing size. If $p \leq 2$, Theorem 1.1(i) gives $s^+(G) > s^-(G)$ and hence $s^+(G) > n+3$. If $p = 4$, contract the four pairwise disjoint triangle blocks to four marked vertices. The remaining blocks are bridges, so the resulting graph is a tree. Deleting three appropriate bridge edges separates the four marked vertices into four components. Thus $V(G)$ is partitioned into four induced triangular unicyclic graphs $G_1, \ldots, G_4$, and

$$s^+(G) \geq \sum_{j=1}^{4} s^+(G_j) > \sum_{j=1}^{4} |V(G_j)| = n.$$

Suppose finally that $p = 3$. Define the *intersection graph* of the triangle blocks to have one vertex for each triangle and an edge when two triangles share a vertex. A *cyclic cluster* is a connected component of this intersection graph. If there is more than one cyclic cluster, then paths between clusters in the block–cut tree contain bridge blocks. Deleting suitable such bridges partitions $V(G)$ into induced connected block graphs, one for each cluster (with any pendant bridge-only branches assigned to the adjacent piece). Each piece has at most three triangle blocks and therefore cycle rank at most three. The odd-unicyclic result, Corollary 4.3, and Proposition 4.4, according to the rank of the piece, show that its positive square energy is strictly larger than its order. Superadditivity then gives $s^+(G) > n$.

We are left with one cyclic cluster. The packing number is the independence number of the triangle intersection graph. Choose three independent vertices, corresponding to pairwise vertex-disjoint triangles T_1, T_2, T_3. The fourth triangle T_0 must meet each T_i, since the intersection graph is connected and there are no edges among the T_i. Thus the intersection graph is $K_{1,3}$. Moreover, the three intersection vertices

$$x_i = V(T_0) \cap V(T_i) \qquad (1 \leq i \leq 3)$$

are distinct: if two were equal, the corresponding petals would intersect. This proves the required one-cluster classification; T_0 is the central triangle and the T_i are its pairwise disjoint petals.

Put

$$U = V(T_1) \cup V(T_2) \cup V(T_3), \qquad F = G - U.$$

The only odd-cardinality collections of pairwise disjoint cycles are the four singleton triangles and the collection of all three petals. Since $(-2\mathrm{i})^3 = 8\mathrm{i}$, the exact Sachs identity for the imaginary part is

$$\operatorname{Im} \Psi_G(t) = -2 \sum_{j=0}^{3} Z_{G-V(T_j)}(t) + 8 Z_F(t). \tag{12}$$

We show that the negative terms dominate pointwise for every $t > 0$.

After deleting T_0, each petal leaves one edge, and these three edges are pairwise disjoint and disjoint from F. Given a matching of F, adjoin an arbitrary subset of these three edges. This is a weight-preserving injection into the matchings of $G - V(T_0)$ and gives

$$Z_{G-V(T_0)}(t) \geq (1 + t^2)^3 Z_F(t) > Z_F(t). \tag{13}$$

Indeed, relative to a monomial of Z_F, choosing r of the three edges contributes t^{6-2r}, whose sum over all choices is $(1+t^2)^3$.

For $i \in \{1,2,3\}$, let H_i be the subgraph induced by $U \setminus V(T_i)$. It has a perfect matching: use the edge $x_j x_k$ of T_0, where $\{i,j,k\} = \{1,2,3\}$, together with the edge joining the two noncentral vertices in each of T_j and T_k. Therefore $Z_{H_i}(t) > 1$: the perfect matching supplies a constant term, and the empty matching supplies the positive term t^6. Taking the union of a matching of H_i and a matching of F gives a weight-preserving product injection into the matchings of $G - V(T_i)$, so

$$Z_{G-V(T_i)}(t) \geq Z_{H_i}(t) Z_F(t) > Z_F(t). \tag{14}$$

In both injections, any additional edges from the displayed subgraphs to the arbitrary forest F are simply ignored; their presence cannot destroy the injected matchings.

Equations (13) and (14) imply

$$\sum_{j=0}^{3} Z_{G-V(T_j)}(t) > 4Z_F(t).$$

Thus (12) is strictly negative for every $t > 0$. The same phase and Coulson argument again yields $s^+(G) > s^-(G)$, and consequently $s^+(G) > |E(G)| = n + 3 > n$. $\square$

The all-triangle conclusion in Proposition 4.5 is also an immediate instance of Corollary 4.2. The direct matching proof above is retained because it proves the stronger asymmetry $s^+(G) > s^-(G)$ in the exceptional packing-three cluster and illustrates how opposite three-cycle phases can nevertheless be dominated.

Proposition 4.6 (Block graphs of cyclomatic number five)**.** *Let G be a connected block graph with n vertices and cyclomatic number* 5. *Then*

$$s^+(G) > n.$$

Thus Conjecture 1.2 of Akbari–Kumar–Mohar–Pragada–Zhang holds strictly for every such block graph.

Proof. The positive clique-block contributions to the cyclomatic number are $1, 3, 6, \ldots$. Hence the cyclic blocks are either five copies of K_3, or one copy Q of K_4 and two copies T_1, T_2 of K_3; all other blocks are bridges. In the first case Corollary 4.2 applies directly.

Consider the second case. Call two cyclic blocks adjacent when they share a cut vertex, and call the connected components under this adjacency *cyclic clusters*. Equivalently, blocks in distinct clusters are separated in the block–cut tree by at least one bridge block. If there is more than one cluster, delete bridge edges on the paths between clusters. This partitions $V(G)$ into vertex sets inducing connected block graphs, one for each cluster, with every bridge-only component assigned to an adjacent piece. Every piece is cyclic, their positive cyclomatic ranks sum to 5, and, because there is more than one piece, each rank is at most 4. The odd-unicyclic theorem, Corollary 4.3, and Propositions 4.4 and 4.5, according to the rank, show that the positive square energy of each piece is strictly larger than its order. Superadditivity then gives $s^+(G) > n$.

It remains that Q, T_1, T_2 form one cyclic cluster. Each external triangle meets Q either at one cut vertex or through a chain containing the other external triangle. In particular, each T_i uses at most one vertex of Q, so at most two vertices of Q lie in $V(T_1) \cup V(T_2)$. Choose

$$z \in V(Q) \setminus \big(V(T_1) \cup V(T_2)\big).$$

Let U consist of z together with all bridge-only branches rooted at z, and put $R = V(G) \setminus U$. Here a bridge-only branch rooted at z is a component attached to the cyclic-block union only at z and containing no cyclic block.

The induced graph $G[U]$ is a tree. This includes the possibility $U = \{z\}$, in which case $s^+(G[U]) = 0 = |U| - 1$. The graph $G[R]$ is connected: $Q - z$ is connected, and the one-cluster incidence among Q, T_1, T_2 remains connected after deleting z, since z lies in neither external triangle; all retained bridge-only branches stay joined at their roots. Its cyclic blocks are precisely the three triangles $Q - z, T_1, T_2$.

Their triangle intersection graph is connected. Indeed, the original one-cluster incidence is connected, and replacing Q by $Q - z$ preserves every incidence with an external triangle because those triangles avoid z. A connected graph on three vertices has independence number at most two, so $\nu_\circ(G[R]) \leq 2$. Theorem 1.1(i) and (1) therefore give

$$s^+(G[R]) > |E(G[R])| = |R| + 2,$$

the equality following because $G[R]$ is connected of cyclomatic number three. Since a tree has equal positive and negative square energies, $s^+(G[U]) = |E(G[U])| = |U| - 1$. Finally,

$$s^+(G) \geq s^+(G[U]) + s^+(G[R]) > (|U| - 1) + (|R| + 2) = n + 1 > n.$$

□

We next state a convenient cactus family. A triangular cactus is a cactus whose cyclic blocks are triangles. A bouquet of triangles is a set of triangle blocks having a common vertex. We say that the triangles of a triangular cactus form at most two bouquets if its set of triangle blocks can be partitioned into at most two subfamilies, each subfamily having a common vertex. Empty subfamilies are permitted.

Corollary 4.7 (One and two bouquets)**.** *Let G be a triangular cactus containing at least one triangle. If its triangles form at most two bouquets, then*

$$s^+(G) > s^-(G).$$

This includes friendship graphs and double friendship graphs, with arbitrary trees attached at their vertices and with arbitrary forest components added.

Proof. All cycles are triangle blocks. A pairwise vertex-disjoint collection contains at most one triangle from each bouquet, and hence has size at most two. Apply Theorem 1.1(i). Tree attachments and forest components create no cycles and do not affect this verification. □

Here a friendship graph means a bouquet of triangles with one common center. A double friendship graph means any triangular cactus whose triangle blocks are the union of two specified bouquets; the two centers may be joined by a path or may coincide. This definition states exactly the property used in Corollary 4.7, rather than relying on a stronger triangle-intersection classification.

The phase-reversed version is equally immediate.

Corollary 4.8. *Let G contain a cycle, and suppose every cyclic block is a cycle of length* $1 \pmod 4$*. If these cycle blocks can be partitioned into at most two bouquets (each subfamily has a common vertex), then $s^+(G) < s^-(G)$.*

Proof. Every cycle is a cyclic block and has length $1 \pmod 4$, while at most one cycle from each bouquet can occur in a vertex-disjoint collection. Thus $\nu_\circ(G) \leq 2$, and Theorem 1.1(ii) applies. □

Remark 4.9 (Why the packing threshold appears)*.* For three vertex-disjoint cycles of the common odd phase, the three-cycle terms in (6) are imaginary and have the opposite sign from the one-cycle terms: $(-2\mathrm{i})^3 = 8\mathrm{i}$ in the $3 \pmod 4$ case and $(2\mathrm{i})^3 = -8\mathrm{i}$ in the $1 \pmod 4$ case. Positivity of the signless matching factors alone therefore no longer fixes the imaginary sign. The argument proves the packing-two theorem sharply as a phase-counting mechanism; it does not assert that every graph with packing number three violates the conclusion.

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